

The figures in the right-hand margin indicate marks for the individual question.

Answers should be concise and entire answer to a question should be together. Give examples and diagrams wherever appropriate. State assumptions wherever made.

1. (a) How is speech represented in a computer in digital form? 3
 (b) What is UTF-8? What are its advantages? 2 + 2
2. (a) What is *inflection* and *derivation* in morphology? Give examples involving affixes. 4
 (b) Why is smoothing required in n-gram based estimates? Why is smoothing also called discounting? 2 + 1
 (c) In applications such as of NLP, *entropy* indicates *number of bits*. What is the physical significance of such an entropy value? 2
3. (a) Briefly describe the *completer* action of Earley's parser. 3
 (b) How is the idea of feature unification useful beyond context free grammar? How is it used? 2 + 2
4. (a) How is the idea of *probabilistic context free grammar* (PCFG) relevant when Earley's parser is used for parsing? How can the probability values of a PCFG be determined? 2 + 2
 (b) Give an example to show how PCFGs fail to deal with *lexical dependencies* (i.e., different probability for different words of a class). 2
5. (a) Justify the need that a meaning representation scheme should be unambiguous as well as allow for vagueness. 2
 (b) Represent the meaning of the following sentence using *First Order Predicate Calculus*- 3
 TU students take dinner at 8 pm in their hostels.
6. (a) Identify various lexical relations (such as, homonymy, synonymy) among the following words- 3
 heat, hit, beat, bit, defeat, success, fit, feat, feet (of a person), feet (plural of unit of 12 inches).
 (b) Give an example to show how a given word can have the thematic role of *experiencer* and *agent*. 2
 (c) Give an example of *metaphor*. 1
7. (a) How can selectional restrictions be used in word sense disambiguation? Give examples of its use and limitation. 4
 (b) How can word sense disambiguation be performed with the help of a lexicon? 3
 (c) Why is stemming of words required in information retrieval? 2
8. (a) How can syntactic constraints help in reference resolution in a discourse? 3
 (b) What is the concept of *recency* in reference resolution? 2
9. (a) What are some advantages and challenges of interlingua based machine translation? 4
 (b) What does *lexical gap* mean? 2